DE–1477 **11**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2015.

First Semester

PRINCIPLES OF INFORMATION TECHNOLOGY

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write about "The new story" of communications and computers. (10)

   (b) What are the various developments in computer technology? Explain. (10)

2. (a) Give a detailed note about the ethics of information technology. (10)

   (b) "The information appliance will change your future". Justify your answer. (10)

3. What are the four types of application software? Explain its features in detail. (20)

4. (a) Explain the features and uses of interface. (10)

   (b) Give a short note on desk top accessories. (10)

5. (a) Explain the concept of electronic data interchange. (10)

   (b) What is ISDN? How is it important? Explain. (10)

6. (a) Explain about the modem and data communication software. (10)

   (b) What are the various factors affecting data transmission? Explain. (10)

7. What is secondary storage devices? Explain any two devices in detail. (20)

8. What is a programming language? Explain the five generations of languages. (20)

DE–1478 **12**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2015.

First Semester

DIGITAL COMPUTER ORGANISATION

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the octal and hexadecimal system in detail. (10)

   (b) Convert the decimal number 157_{10} into binary, octal and hexadecimal. (10)

2. (a) Explain the types of complements and its uses. (10)

   (b) Explain basic theorems and properties of Boolean algebra. (10)

3. (a) Describe the 3 to 8 line decodes with its logic diagram and truth table. (10)

   (b) Explain the 4 bit parallel binary adder in detail. (10)

4. (a) Describe the construction of Arithmetic logic unit in detail. (10)

   (b) Explain the construction of Half adder and full adder with functional table. (10)

5. (a) Explain the shift register in detail with its wave forms. (10)

   (b) Discuss in detail about clocked R-S flip-flops. (10)

6. (a) Describe the static Random access memory with pin diagram. (10)

   (b) Explain about magnetic tapes. (10)

7. (a) Explain the instruction cycle and execution cyle organization of control registers. (10)

   (b) Discuss about DMA. (10)

8. (a) Write in detail about decoders. (10)

   (b) Explain about interrupts and interrupt priorities. (10)

DE–1479 **13**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2015

First Semester

C AND DATA STRUCTURES

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) List out identifiers and keywords and differentiate with examples. (10)

   (b) What is a variable? How to declare various types of variables with examples? (10)

2. (a) Explain various operators available in C with examples. (10)

   (b) Explain with examples about Input/Output statements available in C. (10)

3. (a) List and explain any three control statements with necessary examples. (10)

   (b) Differentiate Switch-Break-Continue statements. (10)

4. (a) Explain functions. How to access function? Give examples. (10)

   (b) List out the library functions. Explain with its uses. (10)

5. (a) Write short notes on arrays; and multidimentional arrays with examples. (10)

   (b) Compare and contract arrays with strings. (10)

6. (a) Explain dynamic memory allocation with example statements. (10)

   (b) What is an array of pointers? Explain with simple example. (10)

7. (a) Explain passing functions to other functions with necessary examples. (10)

   (b) How to create and process a data file? Explain. (10)

8. (a) Write notes on linked list operations. (10)

   (b) Explain trees and binary trees with necessary diagrams. (10)

———————

DE–1480 **14**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2015.

First Semester

OFFICE AUTOMATION

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain about menu commands and toolbars of office application. (10)

   (b) How will you create and open files in MS-Office? Explain in detail. (10)

2. (a) Explain the mail merge with an example. (10)

   (b) How will you set margins in MS-Word? Explain. (10)

3. (a) Explain about paragraph spacing in MS-Word. (10)

   (b) Discuss about tables in MS-Word. (10)

4. (a) What is meant by absolute addresses in MS-Excel formula? Explain with an example. (10)

   (b) Explain number formatting in MS-Excel. (10)

5. Discuss about creating and working with charts in MS-Excel. (20)

6. (a) Discuss about text formatting and alignment in MS-Powerpoint. (10)

   (b) Discuss about handouts in MS-Powerpoint. (10)

7. (a) How will you create a table with primary key in MS-Access? Explain. (10)

   (b) How will you find records in MS-Access table? Explain. (10)

8. (a) Explain about autoforms in MS-Access. (10)

   (b) How will you embed copied data in another document in MS-Office? Explain. (10)

DE–1481 **21**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2015.

Second Semester

OBJECT ORIENTED PROGRAMMING AND C++

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) (i) Discuss in details about object oriented programming paradigm. (5)

   (ii) What are the basic concepts of oops? (5)

   (b) (i) Compare C and C++. (5)

   (ii) What are the fundamentals of C++? (5)

2. (a) Discuss the different types of array with suitable example. (10)

   (b) Explain the different control structures with example. (10)

3. (a) (i) What is an iterative statement? (3)

   (ii) How does a C++ structure different from C++ class? Discuss. (7)

   (b) What is argument passing? Discuss Pass-by-value with an example. (10)

4. (a) Discuss the following with example

   (i) Scope of the variables. (5)

   (ii) Default arguments. (5)

   (b) What is macros and inline function? Discuss in detail with example. (10)

5. (a) Discuss about constructors and destructors in C++ with suitable example. (10)

   (b) What is inheritance? Describe in detail about multiple and hybrid inheritance with example. (10)

6. (a) Describe the following

   (i) Type conversion. (5)

   (ii) Operator overloading. (5)

   (b) Discuss about virtual function and polymorphism with suitable example. (10)

7. (a) Write a C++ program to read the file and write into an another file. (10)

   (b) Discuss in detail about file pointers with suitable example. (10)

8. (a) List the stream classes for console operation. Give example. (10)

   (b) Describe the following :

   (i) End-of-file (5)

   (ii) File direction. (5)

DE–1482 **22**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2015.

Second Semester

SYSTEMS ANALYSIS AND DESIGN

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the characteristics of open systems and man-made information systems. (10)

   (b) Explain the interpersonal and technical skills necessary in system development. (10)

2. (a) Explain the strategies for determining information requirement. (10)

   (b) Discuss about the classification of costs and benefits. (10)

3. (a) Discuss the major requirement of form design. (10)

   (b) Explain HIPO and IPO charts. (10)

4. (a) Explain indexed file organization. (10)

   (b) Explain the various levels of quality assurance. (10)

5. (a) Explain parallel and direct conversion methods. (10)

   (b) Discuss the procedure for hardware and software selection. (10)

6. (a) Explain the decision support system. (10)

   (b) Discuss system development life cycle with prototyping. (10)

7. (a) Write about various interpersonal skills and technical skills that are relevant to systems work. (10)

   (b) Discuss any five steps to be followed in feasibility analysis. (10)

8. (a) Explain different types of interviews and questionnaires. (10)

   (b) Discuss about system flow chart. (10)

———————

DE–1483 **23**

DISTANCE EDUCATION

M.C.A (S) DEGREE EXAMINATION, MAY 2015

Second Semester

VISUAL PROGRAMMING – I

(2005 to 2010 calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Discuss in detail about the overview of developing a visual basic application. (10)

   (b) Explain in detail about the visual basic environment. (10)

2. (a) Discuss in detail about the property window and common properties in visual basic. (10)

   (b) Explain about project explorer and add-ins in visual basic. (10)

3. (a) Compare text box with labels in visual basic with suitable example. (10)

   (b) Explain the steps to creating controls in visual basic. (10)

4. (a) Explain in detail about visual basic decision making statements with example. (10)

   (b) Explain visual basic data types in detail. (10)

5. (a) Explain about procedures and subprocedures in visual basic. (10)

   (b) Write in detail about arrays in visual basic with suitable example. (10)

6. (a) Discuss in detail about list and combobox in visual basic with example. (10)

   (b) Write in detail about events and methods in visual basic with suitable example. (10)

7. (a) Describe the global procedures and variables in visual basic. (10)

   (b) Describe about menus in visual basic with menu editor. (10)

8. (a) Explain sequential file organization in visual basic and create student file. (10)

   (b) Describe about building activex control in visual basic. (10)

———————

DE–1484 **24**

DISTANCE EDUCATION

M.C.A. (S) DEGREE EXAMINATION, MAY 2015.

Second Semester

COMPUTER GRAPHICS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. With neat diagram, explain the working principles of cathode ray tube. (20)

2. Discuss the procedure of generating a circle with midpoint circle generation algorithm. (20)

3. List down the basic two dimensional transformation principles and explain with matrices and derivations. (20)

4. (a) Elucidate the procedure of window-to-viewport coordinate transformation principle. (10)

   (b) Discuss on Sutherland-Hodgeman polygon clipping method. (10)

5. List and explain the different types of three dimensional display methods. (20)

6. Give a neat description on three dimensional projections. (20)

7. (a) Write about Z-buffer algorithm. (8)

   (b) Discuss on user's model that helps to have good user interface design. (12)

8. Explain the role of "Command language and styles of it", in user interface design. (20)

DE–1485 **31**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Third Semester

DISCRETE MATHEMATICS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Construct the truth table for

   (i) $(P \vee Q) \vee \neg P$

   (ii) $P \wedge (P \vee Q)$. (10)

   (b) Show that $(\neg P \wedge (\neg Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$. (10)

2. (a) Obtain the principal conjunctive normal form of the formula $(\neg P \rightarrow R) \wedge (Q \rightleftarrows P)$. (10)

   (b) Show that $S \vee R$ is tautologically implied by $(P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow S)$. (10)

3. (a) Show that for any two sets A and B $A - (A \cap B) = A - B$. (10)

   (b) Given $A = \{2, 3\ 4\}$, $B = \{1, 2\}$, $C = \{4, 5, 6\}$ find $A + B$, $B + C$, $A + C$, $A + B + C$. (10)

4. (a) What are the properties of binary relation? Explain with examples. (10)

(b) Let $X = \{1, 2, 3, 4, 5, 6, 7\}$ and $R = \{< x, y > / x - y$ is divisible by $3\}$. Show that R is an equivalence relation. Draw the graph of R. (10)

5. (a) Let $X = \{1, 2, 3\}$ and f, g, h and s be functions from X to X given by

$f = \{<1, 2>, <2, 3>, <3, 1>\}$

$g = \{<1, 2>, <2, 1>, <3, 3>\}$

$h = \{<1, 1>, <2, 2>, <3, 1>\}$

$s = \{<1, 1>, <2, 2>, <3, 3>\}$

Find $f \circ g$, $g \circ f$, $g \circ h$, $h \circ s$. (10)

(b) Explain the different types of functions with examples. (10)

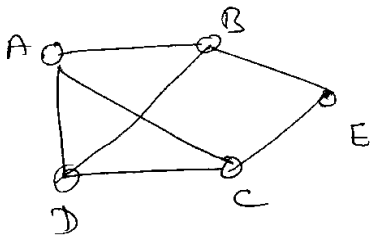
6. (a) Explain about semigroup and monoid with examples. (10)

(b) Show that every finite semigroup has an idempotent. (10)

7. (a) State and prove Lagrange's theorem. (10)

(b) Write a Warshall algorithm to find the path matrix. (10)

8. (a) What do you mean by adjacency matrix for Graph? Find the adjacency matrix for the following graph. (10)



(b) What is a tree? Explain with examples. (10)

DE–1486 **32**

DISTANCE EDUCATION

M.C.A (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Third Semester

OPERATING SYSTEMS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain about system calls. (10)

   (b) Explain about operating system structure. (10)

2. (a) Explain process state, PCB, Process transition and process switch with a neat process state transition diagram. (10)

   (b) Write short notes about Interprocess communication. (10)

3. (a) Explain the methods used to prevent deadlocks. (10)

   (b) Explain about I/O devices. (10)

4. (a) Explain about Mutual exclusion. (10)

   (b) Write short notes about critical sections. (10)

5. (a) Distinguish between

   (i) fixed and variable partition. (5)

   (ii) paging and segmentation. (5)

   (b) Explain various page replacement strategies. (10)

6. (a) Explain about the concept of virtual memory. (10)

   (b) Explain about memory management with linked lists. (10)

7. (a) Discuss the organization of typical file system. (10)

   (b) Write short notes on file sharing concept. (10)

8. (a) Define file and explain various operations on files. (10)

   (b) Explain about directory structure. (10)

DE–1487 **33**

DISTANCE EDUCATION

M.C.A. (S)/ M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Third Semester

RDBMS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What are the advantages of DBMS? (10)

   (b) Explain the different uses Interfaces. (10)

2. (a) Discuss the concepts of Relational Model. (10)

   (b) Write a note on Data Control facilities. (10)

3. (a) Explain the types of databases. (10)

   (b) Discuss about the Oracle Database Architecture. (10)

4. (a) Explain in detail about DML commands. (10)

   (b) How to join multiple tables in a Query? Discuss with example. (10)

5. (a) How to create a table? Describe the table definitions. (10)

   (b) Describe about the languages supported by oracle precompiler. (10)

6. (a) Explain reason for data base objects. (10)

   (b) Explain subqueries, unions and multiple part Queries. Discuss with example. (10)

7. (a) Explain in detail about the client/server systems. (10)

   (b) Explain network system tables with example. (10)

8. (a) Explain the relational database properties with examples. (10)

   (b) What is schema and subschema? Discuss. (10)

———————

DE–1488 **34**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S)(Lateral) DEGREE EXAMINATION, MAY 2015.

Third Semester

UNIX AND SHELL PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is Shell? Explain with example.

   (b) Discuss in detail about inodes.

2. (a) What is meant by filters? Explain any two filters.

   (b) Explain about the *awk* utility in Unix with example.

3. (a) Explain use of looping statement in Shell with example.

   (b) Illustrate the following with example:

      (i) Metacharacters

      (ii) Shell variables.

4. (a) Explain the following shell commands with example.

      (i) trap

      (ii) zap.

   (b) Explain the process of customizing the *cal* command.

5. (a) Explain the following shell commands with example.

   (i) pick

   (ii) news.

   (b) What are the commands used for tracking file changes? Explain.

6. (a) Write a note on standard input and output shell commands.

   (b) What are the string handling functions used in shell? Explain.

7. (a) What is *lex*? Write the overview of *lex*.

   (b) Explain the *ms* formatting commands for document preparation.

8. Write a shell program for function calculator.

DE–1489 **35**

DISTANCE EDUCATION

M.C.A. (S) / M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Third Semester

COMMUNICATION SKILLS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Define communication and explain the process of communication with the help of schematic representation. (10)

   (b) Describe about barriers to communication and classification of barriers. (10)

2. (a) Explain about the reading speed and what are the reading techniques to be followed? (10)

   (b) What are the guidelines for preparing an agenda and the importance of writing minutes? (10)

3. (a) Discuss the varied purposes of writing the business letters and explain the various courtesy titles. (10)

   (b) What are the types of listening and about the traits of a good listener? (10)

4. (a) What are the basics of telephone communication? (10)

   (b) What do you mean by brain storming and explain the types of brainstorming technique? (10)

5. (a) Explain the purpose and process of group discussion. (10)

   (b) What are the four components generally evaluated in a group discussion? (10)

6. (a) Why body language is considered as a powerful communication? (10)

   (b) Explain the importance and techniques for negotiation. (10)

7. (a) Explain in detail the meaning, objectives and importance of meetings. (10)

   (b) Write about mock meetings and seminars. (10)

8. (a) Discuss the importance and methods of note making. (10)

   (b) Describe the various techniques for delivering presentation. (10)

DE–1490 **41**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A.(S) (Lateral) DEGREE EXAMINATION, MAY 2015

Fourth Semester

COMPUTER NETWORKS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain transmission media with necessary diagrams. (12)

   (b) Write short notes on telephone system. (8)

2. (a) Compare and contrast ISDN and ATM (10)

   (b) Write notes on communication satellites. (10)

3. (a) Explain sliding window protocols. (12)

   (b) Describe finite state models. (8)

4. (a) Discuss about Petri Net models. (10)

   (b) Explain ALOHA. (10)

5. (a) Write down the network layer design issues and its functions. (12)

   (b) Define Routing and explain any two routing techniques with algorithms. (8)

6. (a) What is subnet? Explain. (10)

   (b) Describe the features of ARP and RARP. (10)

7. (a) Explain the elements of transport protocols. (10)

   (b) Write short notes on multiplexing. (10)

8. (a) Explain the importance of network security. (10)

   (b) Explain the need of WWW. (10)

DE–1491 **42**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Fourth Semester

ACCOUNTING AND FINANCIAL MANAGEMENT

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) State and explain the rules of accounts with suitable examples.

   (b) What is Trial Balance? Explain its objectives and uses.

2. (a) What is Break-Even-Analysis? Explain its assumptions and uses.

   (b) What are the objectives and advantages of Budgeting.

3. The following ledger balances were extracted from the books you are required to prepare trading and profit and loss accounts and balance sheet as on 31st March 2008.

Debit balance	Rs.	Credit balance	Rs.
Drawings	2,000	Capital	10,000
Purchases	20,800	Sales	27,500
Stock on 1.4.2007	6,900	Creditors	8,100
Rent	1,000	Discount	270
Furniture	900	Return outwards	580

Debit balance	Rs.	Credit balance	Rs.
Machinery	5,000	Bank loan	3,000
Travelling expenses	650		
Bad debts	120		
Return inwards	300		
Carriage inwards	400		
Wages	325		
Salaries	900		
Interest	480		
Carriage outwards	700		
Insurance	900		
Cash in hand	575		
Debtors	7,500		
	49,450		49,450

Adjustments:

(a) Closing stock was valued at Rs. 8900

(b) Insurance prepaid Rs. 250

(c) Outstanding salaries Rs. 100 and outstanding rent Rs. 200

(d) Interest on bank loan Rs. 150

(e) Depreciate machinery and furniture by 10% each

(f) Provide for doubtful debts at 5% on debtors.

4. Bala Ltd., supplies you the following balance sheet on 31. 12.2011 and 31.12.2012.

Liabilities	2011	2012	Assets	2011	2012
	Rs.	Rs.		Rs.	Rs.
Share capital	70,000	74,000	Bank balance	9,000	7,800
Bonds	12,000	6,000	Accounts		
A/c payable	10,360	11,840	Reveivable	14,900	17,700
Provision for			Inventories	49,200	42,700
Doubtful debts	700	800	Land	20,000	30,000
Reserves and			Goodwill	10,000	5,000
Surplus	10,040	10,560			
	1,03,100	1,03,200		1,03,100	1,03,200

Following additional information has also been supplied to you:

(a) Dividends amounting to Rs. 3,500 were paid during the year 2012.

(b) Land was purchased for Rs. 10,000.

(c) Rs. 5,000 were written off on goodwill during the year.

(d) Bonds of Rs.6,000 were paid during the course of the year.

You are required to prepare a Cash Flow Statement.

5.

Debtor's velocity	3 months
Creditors' velocity	2 months
Stock velocity	8 times
Capital-turnover ratio	2.5 times
Fixed assets - turnover ratio	8 times
Gross profit turnover ratio	25%

Gross profit in a year amount to Rs. 80,000. There is no long term loan or overdraft. Reserves and Surplus amount to Rs. 28,000. Liquid assets are Rs. 97,333.

Closing stock of the year is Rs. 2,000 more than the opening stock. Bills receivable amount to Rs. 5,000 and Bills payable to Rs. 2,000

Find out

(a) Sales

(b) Sundry debtors

(c) Closing stock

(d) Sundry creditors

(e) Fixed assets

(f) Proprietors' fund

Make out the Balance Sheet with as many details as possible.

6. Raj Corpn. Ltd. has prepared the following budget estimates for the year 1999-2000

	Rs.
Sales (units)	15,000
Fixed Expenses	34,000
Sales	1,50,000
Variable costs	6 per unit

You are required to:

(a) Find the p/v ratio, break - even point and margin of safety.

(b) Calculate the revised P/V ratio, break - even point and margin of safety in each of the following cases:

(i) Decrease of 10% in selling price

(ii) Increase of 10% in variable costs

(iii) Increase of sales volume by 2,000 units

(iv) Increase of Rs. 6,000 in fixed costs.

7. From the following particulars, prepare Stores Ledger showing the issue of materials under FIFO method:

01.11.2006	Purchased 200 units @ Rs. 200 per unit
04.11.2006	Issued 150 units
06.11.2006	Purchased 200 units @ Rs. 200 per unit
10.11.2006	Issued 100 units
17.11.2006	Purchased 200 units @ Rs. 210 per units
18.11.2006	Issued 220 units
24.11.2006	Purchased 150 units @ Rs. 230 per units
25.11.2006	Issued 190 units
28.11.2006	Issued 30 units
30.11.2006	Purchased 100 units @ Rs. 220 per unit

8. A company expenses to have Rs. 37,500 cash in hand on 1st April 2012 and requires you to prepare an estimate of cash position during the three month April to June 2012. The following information is supplied to you:

Month	Sales	Purchases	Wages	Factory expenses	Office expenses	Selling expenses
	Rs.	Rs.	Rs.	Rs.	Rs.	Rs.
February	75,000	45,000	9,000	7,500	6,000	4,500
March	84,000	48,000	9,750	8,250	6,000	4,500
April	90,000	52,000	10,500	9,000	6,000	5,250
May	1,20,000	60,000	13,500	11,250	6,000	6,570
June	1,35,000	60,000	14,250	14,000	7,000	7,000

Other information:

(a) Period of credit allowed by suppliers - 2 months.

(b) 20% of sales are for cash and period of credit allowed to customers for credit sales is one month.

(c) Delay in payment of wages and all expenses - 1 month

(d) Income tax of Rs. 57,000 is due to be paid on June 15,2012

(e) The company is to pay dividends to shareholders and bonus to workers of Rs. 15,000 and Rs. 22,500 respectively in the month of April.

(f) Plant has been ordered and is expected to be received and paid in May. It will cost Rs. 1,20,000.

DE–1492 **43**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Fourth Semester

RESOURCE MANAGEMENT TECHNIQUES

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) What is O.R? Write the mathematical model of L.P.P. (5)

   (b) Solve the following LPP by dual simplex method. (15)

   Minimize $z = 2x_1 + x_2$

   Subject to :

   $3x_1 + x_2 \geq 3$

   $4x_1 + 3x_2 \geq 6$

   $x_1 + 2x_2 \leq 3$ and $x_1, x_2 \geq 0$.

2. (a) What is an assignment problem? Write the LP model of assignment problem. (5)

   (b) Find the optimum solution to the following T.P. (15)

		To					
		A	B	C	D	E	Supply
	1	8	10	12	17	15	100
	2	15	13	18	11	9	150
From	3	14	20	6	10	13	160
	4	13	19	7	6	12	280
Demand		70	170	50	210	190	

3. (a) Write the operational characteristics of queuing system. (10)

(b) The cars arrive at a service station is 5 per hour and it can be serviced at the rate of 8 per hour in Poisson distribution. Find

(i) Average number of cars in the system

(ii) Average number of cars in the queue

(iii) Average time of a car in the system

(iv) Average time of a car in the queue. (10)

4. (a) Discuss about various random numbers generation methods. (10)

(b) Write the algorithm for single queue and single server queuing system. (10)

5. (a) Explain about various costs involved in the inventory. (8)

(b) A company has a demand of 12,000 units per year for a certain item and it can produce 2000 units per month of such items. The cost of one setup is Rs. 400 and the holding cost/unit/month is Re. 0.15. Find

(i) The economic order quantity

(ii) The maximum inventory

(iii) The total cost per year if the cost of one unit is Rs. 4.00

(iv) The manufacturing time. (12)

6. (a) Write down the rules for construction of network diagram. (8)

(b)

Activity :	1-2	1-3	2-3	2-4	3-4	4-5
Duration :	20	25	10	12	5	10

(i) Draw the network diagram

(ii) Compute various time estimates

(iii) Find critical path and total duration. (12)

7. Machine A costs Rs. 45,000 and the operating costs are estimated Rs. 1,000 for the first year and then increasing by Rs. 10,000 per year in the 2nd and subsequent years. Machine B costs Rs. 50,000 and operating costs are Rs. 2,000 for the first year and then increasing by Rs. 4,000 per year in the second and subsequent years. If you have a machine of type A, should you replace it with B? If so, when? Assume that both the machines prove to resale value, and that the future costs are not considered. (20)

8. The following failure rates have been observed for a certain type of transistors in a computer.

Weeks :	1	2	3	4
Prob. of failure :	0.05	0.13	0.25	0.43
Weeks :	5	6	7	8
Prob. of failure :	0.68	0.88	0.96	1.00

There are 1000 transistors in use. The cost of replacing an individual failed transistor is Rs. 1.25 and the cost of group replacement is 30 paise per transistor. Determine both individual and group replacement cost. Suggest best policy to be adopted. (20)

DE–1493 **44**

DISTANCE EDUCATION

M.C.A.(S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Fourth Semester

OBJECT ORIENTED ANALYSIS AND DESIGN

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain the elements of object model. (10)

   (b) How does object oriented approach differ from traditional approach in system development. (10)

2. What is an object? Explain the various kinds of relationship among objects with an example. (20)

3. Define the nature of a class. Explain the various relationships among classes, with a relevant example. (20)

4. (a) What is classification? Explain the importance of proper classification. (10)

   (b) Describe the general approaches to classification. (10)

5. Briefly explain the major phases of unified process in UML. (20)

6. Explain with an example, how use case modeling is used to describe functional requirements. Identify the actors, scenario and use cases for the example. (20)

7. Explain the structure of class diagram. Draw class diagram for ATM operation. (20)

8. (a) How is activity diagram used to model an operation? (10)

   (b) Enumerate the steps to model a work flow with reference to activity diagrams (10)

DE–1494 **45**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Fourth Semester

INTERNET PROGRAMMING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain security and encryption standards in detail. (10)

   (b) What is WWW? Explain the features of internet. (10)

2. (a) Explain the features of perl. (10)

   (b) Discuss the functionality of CGI. (10)

3. (a) Explain the importance of Java in internet programming. (10)

   (b) Write short notes on scripting languages. (10)

4. (a) List the HTML tags for formatting and frame design. (12)

   (b) Discuss on shockwave and lingo. (8)

5. (a) How do create an active x control to activate a web page? (12)

   (b) Discuss about VDO live technology. (8)

6. (a) Write in details about images and hyperlinks. (12)

   (b) How to create a custom integrated application with multi protocols? (8)

7. (a) Discuss about web graphics. (8)

   (b) Write short notes on site and page design concepts. (12)

8. (a) Discuss the importance of animation in internet programming. (12)

   (b) Write short notes on dynamic graphics. (8)

———————

DE–1495 **51**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Fifth Semester

IMAGE PROCESSING AND ANALYSIS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Describe in detail about brightness adaptation and contrast. (10)

   (b) Explain the components of image processing system. (10)

2. (a) Write a detailed note on low-pass filtering. (10)

   (b) Discuss about image averaging in detail. (10)

3. (a) Explain in detail about Lossy compression methods. (10)

   (b) Describe the Huffman coding technique with an example. (10)

4. (a) Discuss in detail about threshold selection methods. (10)

   (b) Write a note on edge detection. (10)

5. (a) Describe about topological attributes. (10)

   (b) Explain in detail about graph matching. (10)

6. (a) Explain in detail about statistical classification. (10)

   (b) Discuss about edge elements extraction by thresholding. (10)

7. (a) Give a detailed note on quad tree coding. (10)

   (b) Explain about restoration by homomorphic filtering. (10)

8. (a) Describe the elements of digital geometry. (10)

   (b) Write a note on photometric model. (10)

———————

DE–1496 **52**

DISTANCE EDUCATION

M.C.A.(S)M.C.A.(S)(LATERAL) DEGREE EXAMINATION, MAY 2015.

Fifth Semester

SOFTWARE ENGINEERING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain about the various software size factors.

   (b) Write a detailed note on spiral model. List out its advantages and disadvantages.

2. (a) Justify the role of management in software development.

   (b) Discuss in detail about the process and project metrics.

3. (a) List the different cost factors involved in software and explain.

   (b) How to plan an organizational structure? Explain.

4. (a) Explain the following cost estimation techniques.

   (i) Delphi cost estimation

   (ii) Constructive cost model.

   (b) Enumerate the different software resources and explain.

5. (a) Describe the process of software scheduling.

   (b) List and explain the different risks involved in software.

6. (a) Write about the software prototyping in detail.

   (b) Briefly explain about the software specification techniques.

7. (a) Explain in detail about the major two module level design concepts of a software.

   (b) Discuss in detail about the Basis Path Testing.

8. (a) Write in detail about the interface design and specify the guidelines.

   (b) Explain the importance of the following testing strategies.

      (i) validation testing

      (ii) system testing.

DE–1497 **53**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (Lateral) DEGREE EXAMINATION,
MAY 2015

Fifth Semester

VISUAL PROGRAMMING – II

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain about Hungarian notation. (10)

   (b) What are windows messages? Explain. (10)

2. (a) Describe mainframe object with an example. (10)

   (b) What are mouse events? Explain. (10)

3. (a) How do you add shortcut keys and accelerator keys? (10)

   (b) How to add menus and connect menu items to your programs? (10)

4. (a) Discuss in detail about message boxes. (10)

   (b) Write a VC++ program to implement list and combo boxes. (10)

5. (a) Write a VC++ program to draw any three geometric shapes. (10)

(b) Explain the following :

(i) Serialization (5)

(ii) Splitter windows. (5)

6. (a) Write note on sequential and random access files. (10)

(b) What is meant by exception handling? Give an example. (10)

7. (a) Explain the technique of access database. (10)

(b) Discuss about building Active X container application. (10)

8. (a) Write a VC++ program to create fibanoci series for n numbers. (10)

(b) Explain the following :

(i) Break points (5)

(ii) View object. (5)

DE–1498 **54**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Fifth Semester

COMPILER DESIGN

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss about the compiler and translators. (10)

   (b) Explain the structure of a compiler in detail. (10)

2. (a) Explain the implementation of lexical analyser in detail. (10)

   (b) Discuss about the regular expression with example. (10)

3. (a) Explain shift-reduce parsing in detail. (10)

   (b) Explain derivation and parse trees in detail. (10)

4. (a) Explain top-down parsing in detail. (10)

   (b) Explain operator precedence parsing in detail. (10)

5. (a) Explain the implementation of syntax directed translator in detail. (10)

   (b) Write in detail about postfix notations and post fix translations with example. (10)

6. (a) Explain lexical phase error in detail. (10)

   (b) Explain error detection and error recovery in detail. (10)

7. (a) Discuss about the syntactic-phase errors with examples. (10)

   (b) Explain the representation of scope information in detail. (10)

8. (a) Explain simple code generator in detail. (10)

   (b) Write and explain the algorithm to construct a DAG in detail. (10)

DE–1499 **55**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Fifth Semester

DATA WAREHOUSING AND MINING

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss about Information and production factor. (10)

   (b) Write in detail about query tools. (10)

2. (a) Write short notes on self learning computer systems. (10)

   (b) Describe the data cleaning process. (10)

3. (a) Explain the OLAP tools. (10)

   (b) Write about Genetic algorithm in detail. (10)

4. (a) Explain the KDD environment. (10)

   (b) Explain the architecture of data warehouse with diagram. (10)

5. (a) Explain in detail about partitioning strategy (10)

   (b) Describe the data marting and Meta data. (10)

6. (a) Discuss about Hardware architecture of data warehouse. (10)

   (b) Write in detail about service level agreement. (10)

7. (a) Describe in detail about backup and recovery in data warehousing. (10)

   (b) Write notes on capacity planning. (10)

8. (a) Explain about testing the data warehouses. (10)

   (b) Explain the features of data warehouse in detail. (10)

———————

DE–1500 **61**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Sixth Semester

SOFTWARE PROJECT MANAGEMENT

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss the steps to be followed in preparing software projects.

   (b) Explain the problems with software projects.

2. (a) Describe the COCOMO cost model.

   (b) Explain the objectives of activity planning.

3. (a) Explain any three methods by which the progress of a project is visualized.

   (b) Discuss the forward pass calculation in a network with an example.

4. Give an overview of risk management.

5. Write the issues related to resource allocation.

6. (a) Elaborate the methods used for identifying the activities of a project.

   (b) How are prototypes classified? Explain.

7. Describe the functions and responsibilities of software configuration management.

8. Explain the following activities in PRINCE

   (a) Standing up a project

   (b) Imitating the project

   (c) Controlling a stage.

DE–1501 **62**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Sixth Semester

MULTIMEDIA SYSTEMS

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) What is Multimedia? Describe the uses of Multimedia. (10)

   (b) Explain the minimum configuration for MPC level 1 and MPC level 2. (10)

2. (a) Discuss the features of any two commercial development tools. (10)

   (b) Write a short note on Open Document Architecture. (10)

3. (a) Describe the various applications of using Image Media. (10)

   (b) Compare Analog and Digital Video. (10)

4. (a) What is an animation? Explain. (10)

   (b) What are the attributes of a block of text? Explain. (10)

5. (a) Explain the parameters for evaluating a compression system. (10)

   (b) Discuss about the JPEG Compression Techniques. (10)

6. (a) What is DVI? Explain. (10)

   (b) Write a short note on Object Oriented Multimedia. (10)

7. (a) Explain about the transform and format classes of object oriented multimedia. (10)

   (b) What are the factors influencing the choice of a suitable storage system for multimedia? Describe. (10)

8. (a) What are the types of Compact Disk? Explain. (10)

   (b) Explain the applications of multimedia in training and education. (10)

———————

DE–1502 **63**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015.

Sixth Semester

NEURAL NETWORKS

(2005 to 2010 Calendar year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the properties of neural networks.

   (b) Explain the basic types of activation function.

2. (a) Discuss about applications of artificial neural networks.

   (b) Discuss about biological neural networks.

3. (a) List and explain any two types of learning process.

   (b) Discuss about the learning tasks.

4. (a) What is Boltzmann machine? Explain.

   (b) What is meant by correlation matrix memory? Explain.

5. (a) Explain about the perception convergence theorem.

   (b) Explain about the architectural graph of a multilayer perceptions.

6. (a) Compare Hop field network and BAM.

   (b) Explain the stable states of the discrete and continuous various of the Hop field Model.

7. (a) Explain the Adaptive Reasoning Theory architecture.

   (b) Discuss about the types of ART.

8. (a) Write a note on optical neural network.

   (b) Write a note on unsupervised artificial neural networks.

DE–1503 **64**

DISTANCE EDUCATION

M.C.A. (S)/M.C.A. (S) (Lateral) DEGREE EXAMINATION, MAY 2015

Sixth Semester

C SHARP (C#)

(2005 to 2010 Calendar Year)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss the necessity of common language runtime and class library. (10)

   (b) Write a note on error handling. (10)

2. (a) What are the features of C#? (10)

   (b) Write a note on metadata and jiters. (10)

3. (a) How to define methods in C#? (10)

   (b) Discuss the concept of events in C#. (10)

4. (a) Explain control statements with example. (10)

   (b) Write a detailed note on namesapce. (10)

5. (a) How to prepare a documentation? (10)

   (b) Write a note on conditional compilation. (10)

6. (a) Explain datatypes in C#. (10)

   (b) Explain constructors and destructors with examples. (10)

7. (a) How to set the break points? (10)

   (b) Describe intermediate language disassembler. (10)

8. (a) Discuss about struct and enumeration. (10)

   (b) Explain boxing and unboxing. (10)